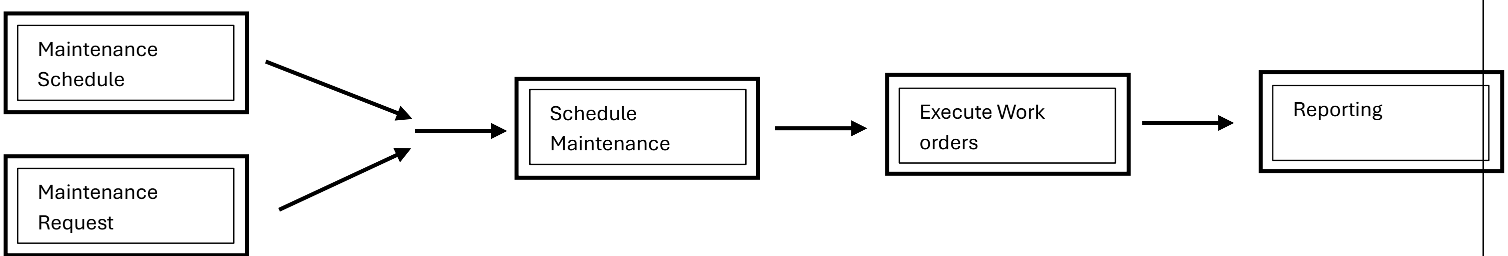
Asset Management: Maintenance Overview

Maintenance Process Flow

When an asset requires attention, whether due to a fault, condition monitoring, or scheduled servicing—a maintenance request is created. This request is reviewed and, upon approval, converted into a work order that groups one or more maintenance jobs

For preventive maintenance, D365 FO supports time-based and usage-based maintenance plans that can automatically generate work orders based on defined criteria. Throughout the process, the system enables comprehensive tracking, reporting, and analysis of asset performance and maintenance efficiency



Base Setups

Maintenance Job type categories

Maintenance Job type

Maintenance Job type variants

Maintenance Job trade

Maintenance Job type defaults

Navigation for all the above forms are defined below

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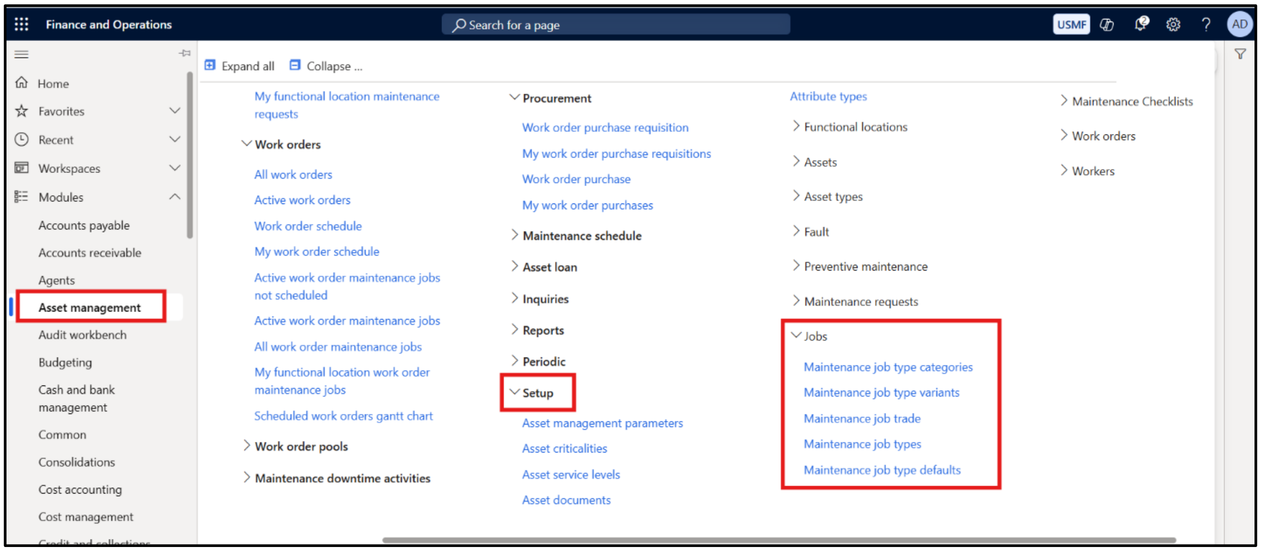
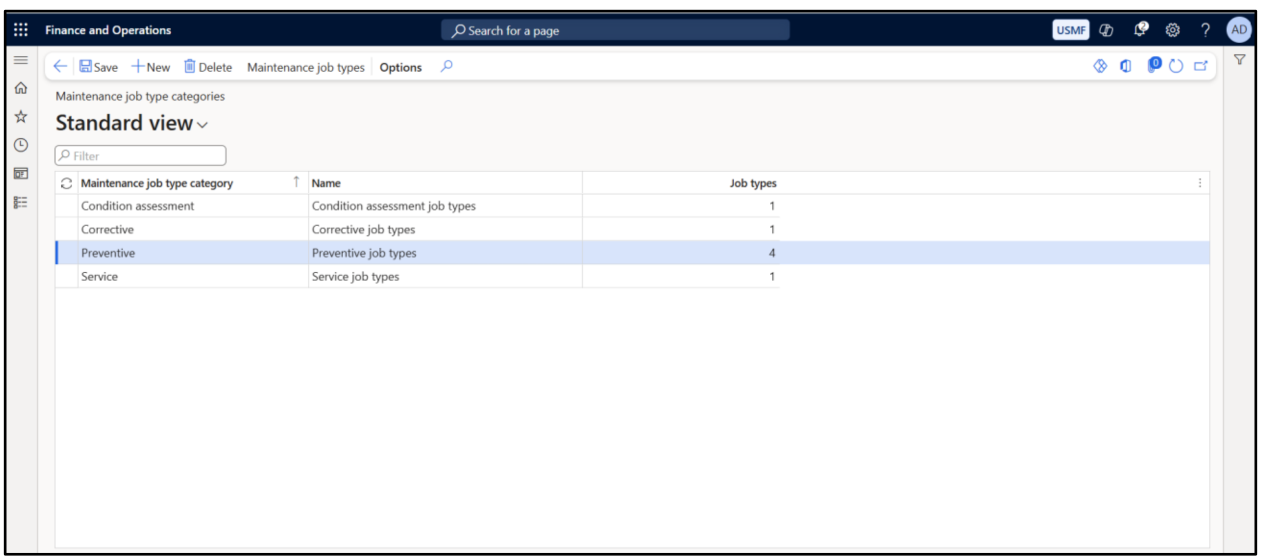


- <u>**Maintenance Job type categories**</u>

These categories help in standardizing maintenance tasks and enabling efficient scheduling, tracking, and reporting. It can be understood as a Job group based on certain common criteria.

<u>Navigation</u>: Asset Management > Setup > Jobs > Maintenance Job type categories



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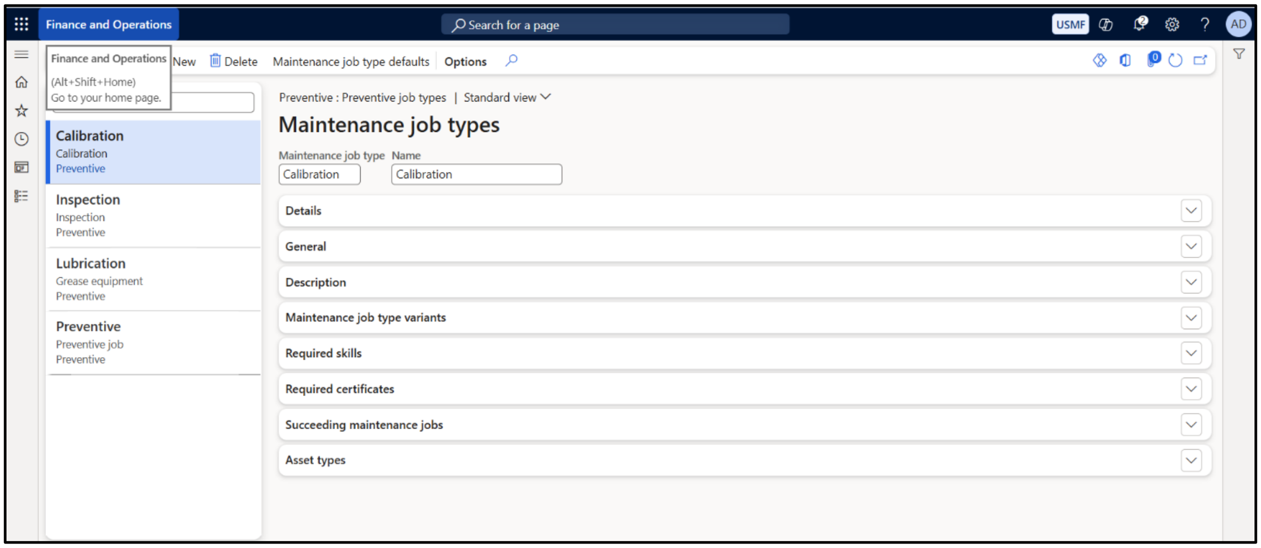
Each category is assigned or associated with a job type.

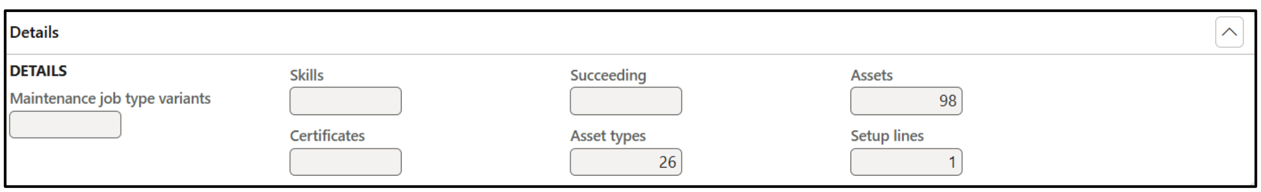
- **<u>Maintenance Job type</u>**

Maintenance Job Types are used to categorize and define the nature of maintenance tasks. These job types help standardize maintenance activities, making planning, scheduling, and tracking more efficient.

<u>Navigation</u>: Asset Management > Setup > Jobs > Maintenance Job type



Each Maintenance Job type is associated with a Category.

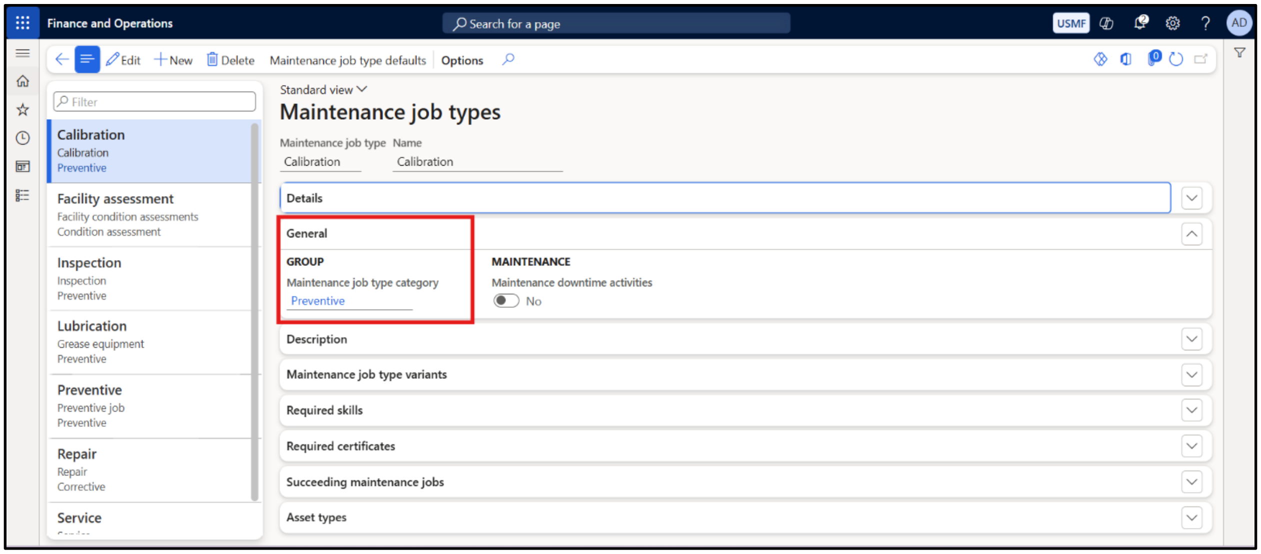


The Details Fast tab contains a number of variants, Skills (required to perform the task), Certificates required, any succeeding jobs, Asset types and Assets associated with this Job. All these can be detailed on this form

We will be discussing the necessary fast tabs below:

1. **General**
2. **Maintenance job type variants**

General

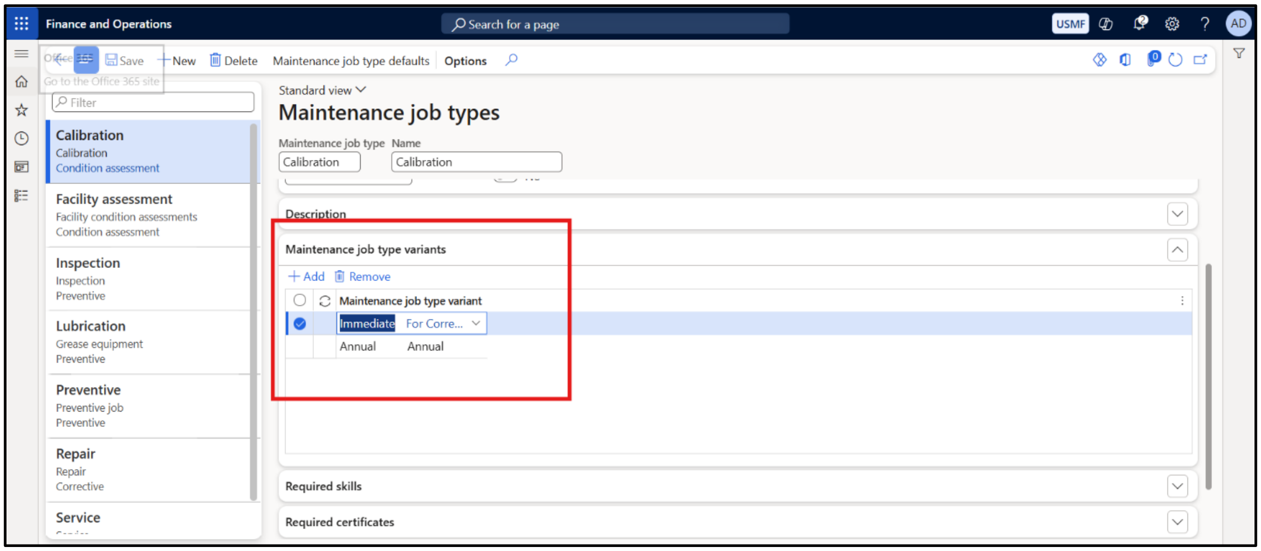


The maintenance job type needs to be mapped to the maintenance job type category. This is a mandatory requirement.

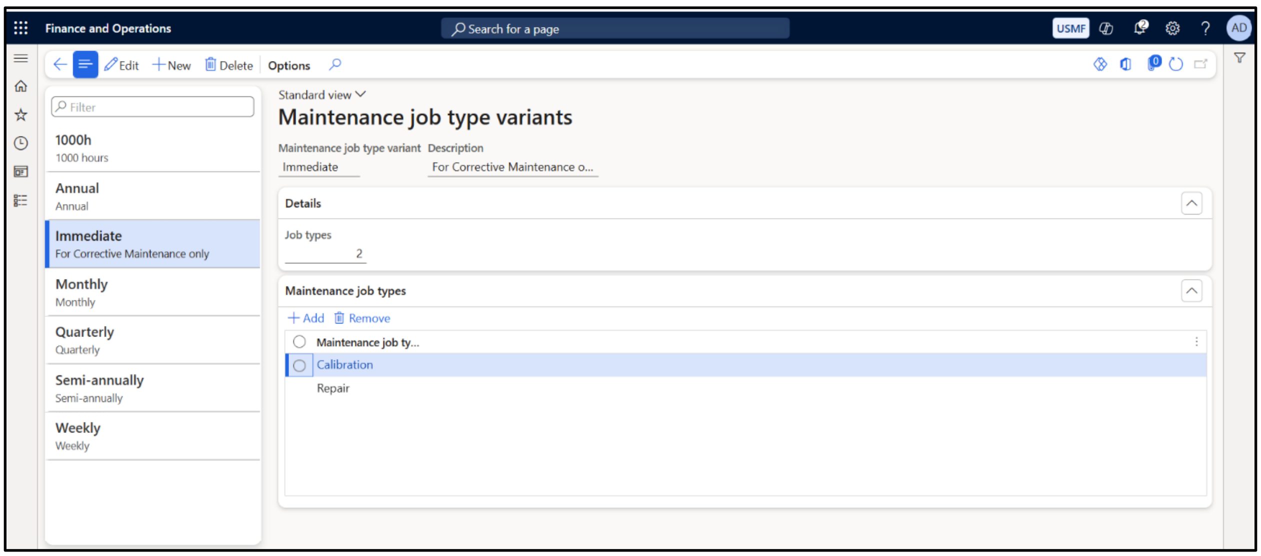
If the maintenance job requires machines to be non-functional, the maintenance downtime activities toggle need to be enabled

- **Maintenance job type variants**

Job Type Variants are sub-classifications or specific versions of a broader Job Type. While a Job Type describes the general nature of maintenance (e.g., “Inspection”), the Variant defines the *specific procedure or context* (e.g., “Monthly Electrical Inspection” vs. “Quarterly Mechanical Inspection”)



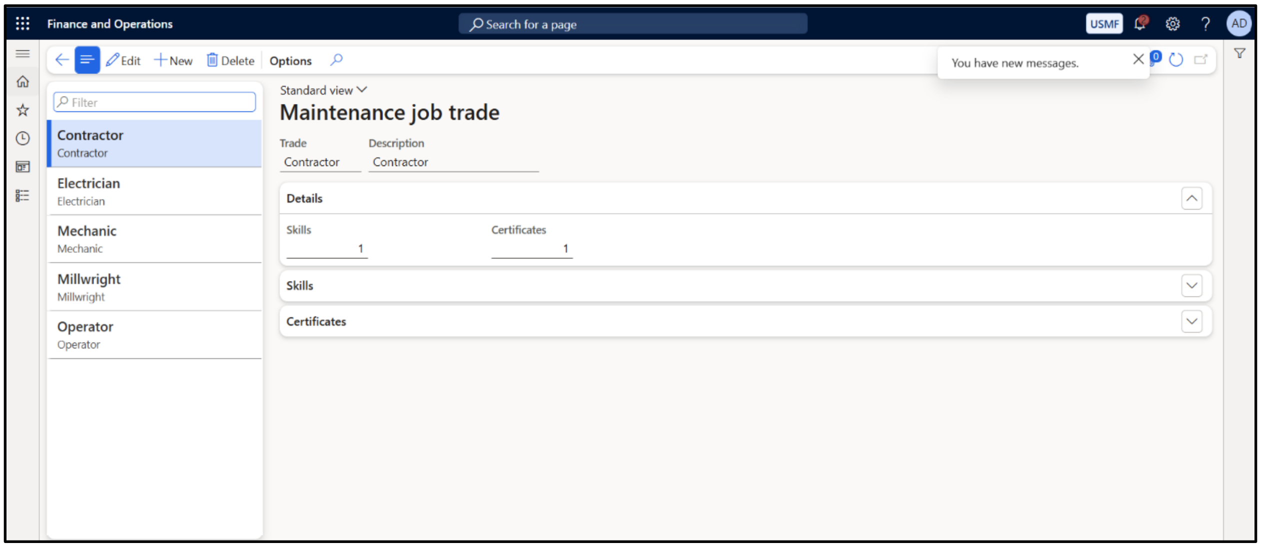
This needs to be mapped to Maintenance job types. This is an optional setup. To create new variants, navigate to **Asset Management > Setup > Jobs > Maintenance Job type** variants and create different variants which can then be mapped to maintenance job types



- <u>**Maintenance Job trade**</u>

Maintenance Job Trade refers to the trade or skill set, and certifications are required to perform a specific maintenance task. It helps in assigning the right resources (technicians or teams) based on their expertise. This is an optional setup.

<u>Navigation</u>: Asset Management > Setup > Jobs > Maintenance Job trade



These can then be mapped to Maintenance job types

- **Maintenance Job type defaults**

Like Functional location type defaults, Asset type defaults, we also have maintenance job type defaults. As the name suggests, certain fields should always be defaulted when a particular Maintenance job type is selected.

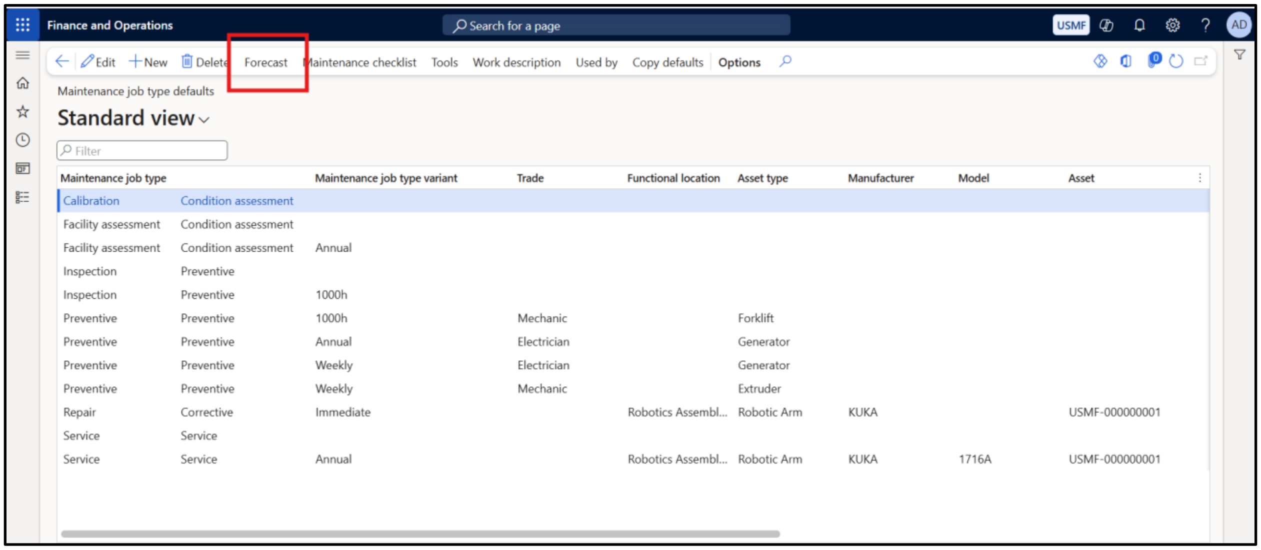
They typically include:

- **Default job trade**
- **Default job variant**
- **Functional location**
- **Default asset type**
- **Resource requirements (e.g., workers, tools)**
- **Spare parts/materials**
- **Checklist templates**
- **Forecast**

This is the first time we have seen an integration of Asset Management with Project Management and Accounting.

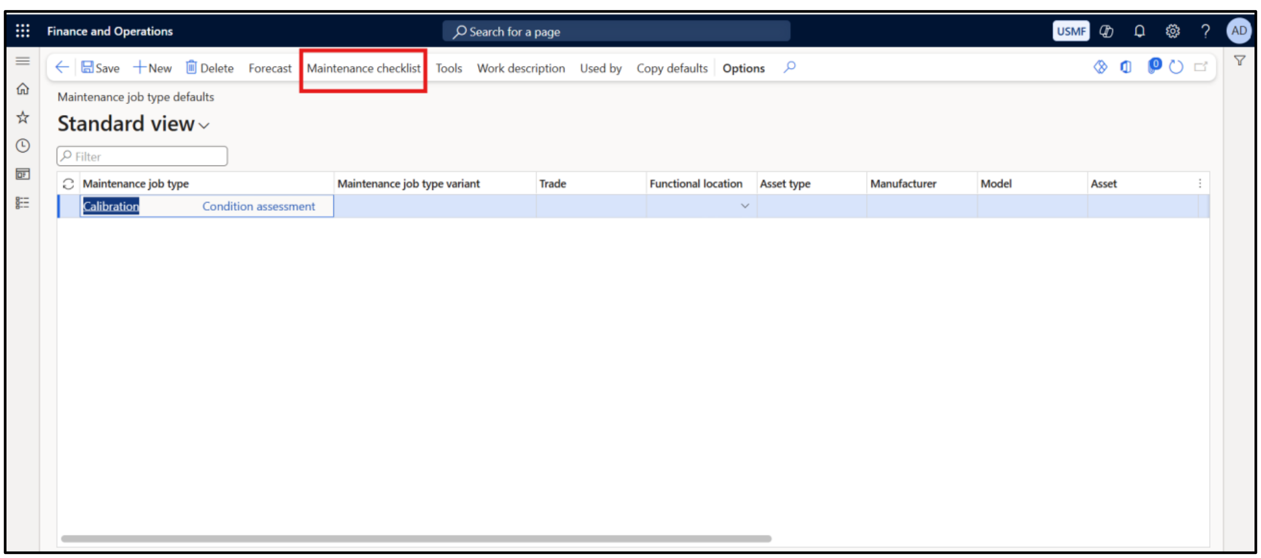
Let us see it more in Details:

- **<u>Forecast</u>**

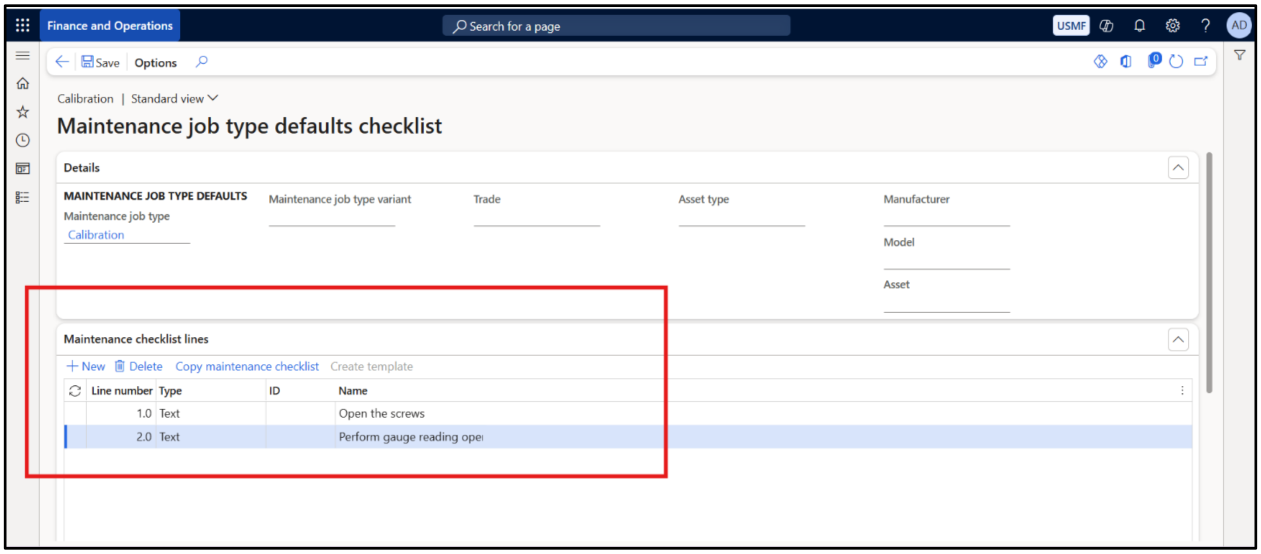


The Forecast on the Maintenance job type defaults can be used to store the **Hour forecast** and **Item forecast**. These forecasts can be directly used as consumption values for posting **financial transactions** on the **workorder.** This would be explained through an example in the next document when we discuss workorder in details.

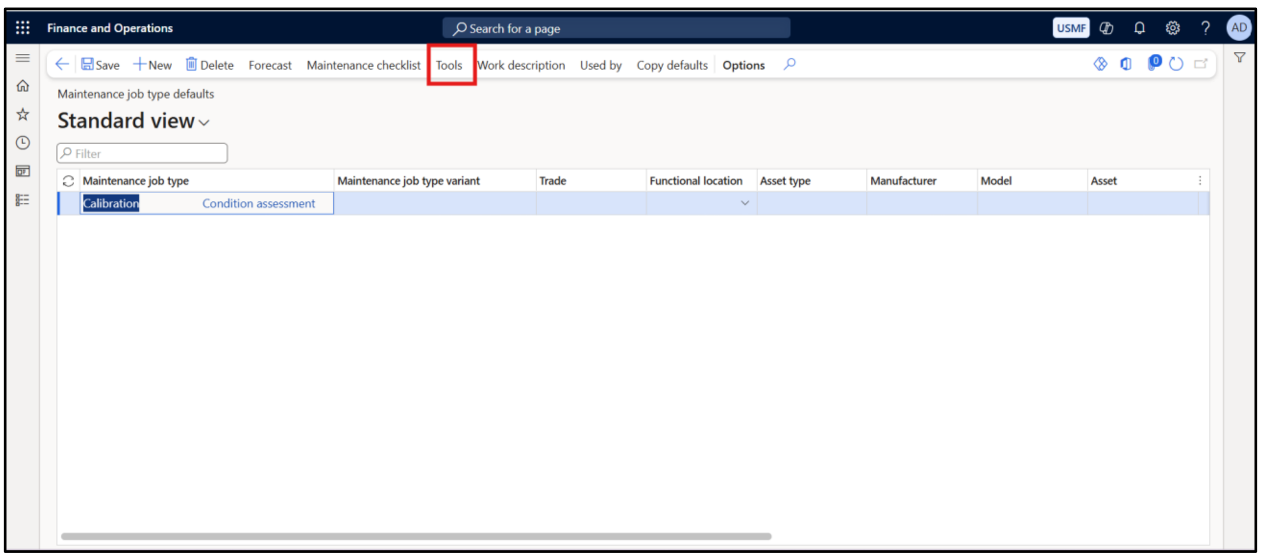
- **<u>Maintenance Checklist</u>**



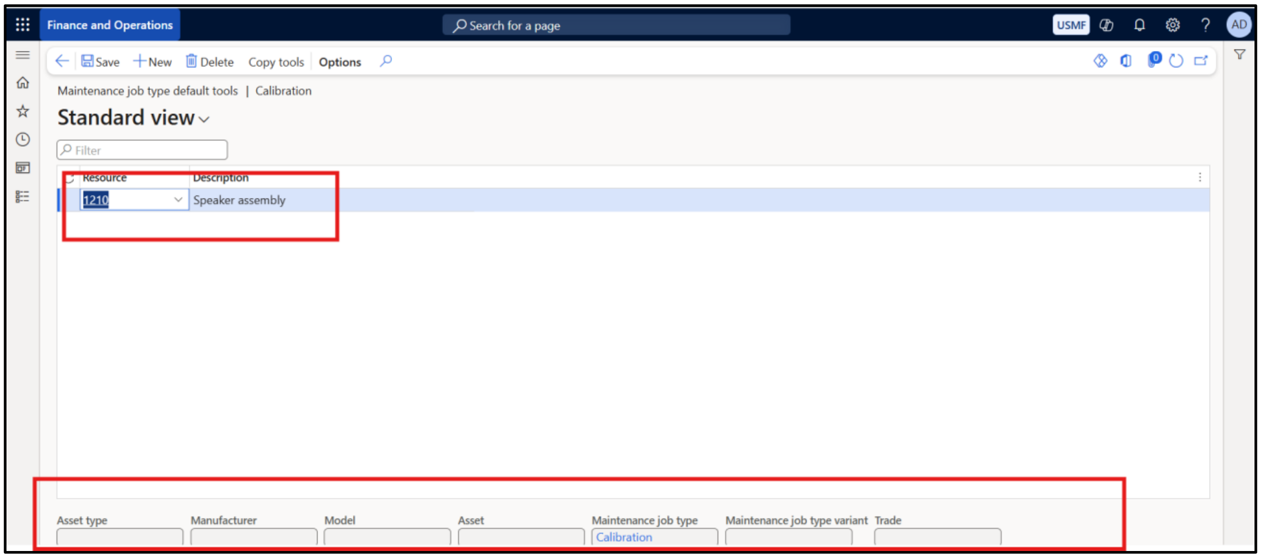
This contains the set of instructions which must be followed by a worker when executing the Workorder.



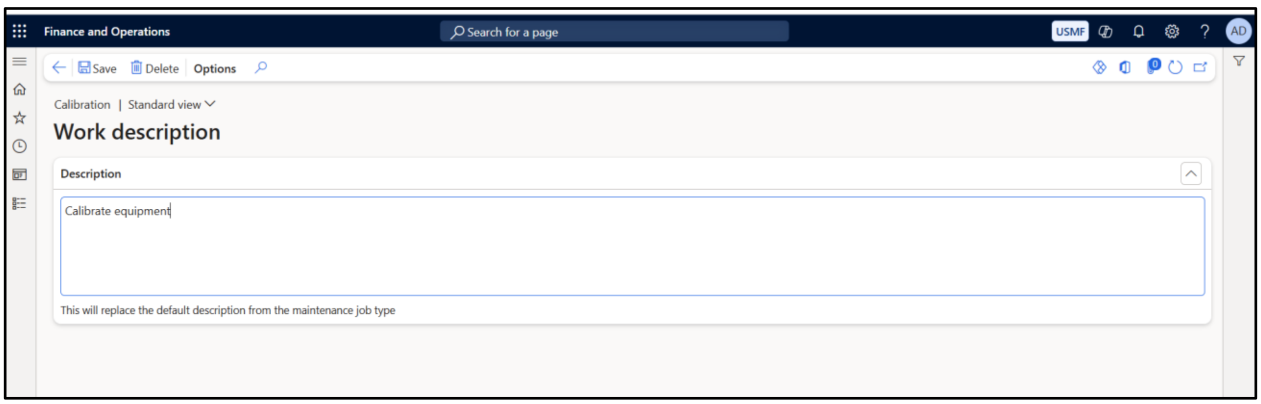
- **<u>Tools</u>**



These are used to associate Resource groups with a particular Maintenance job type

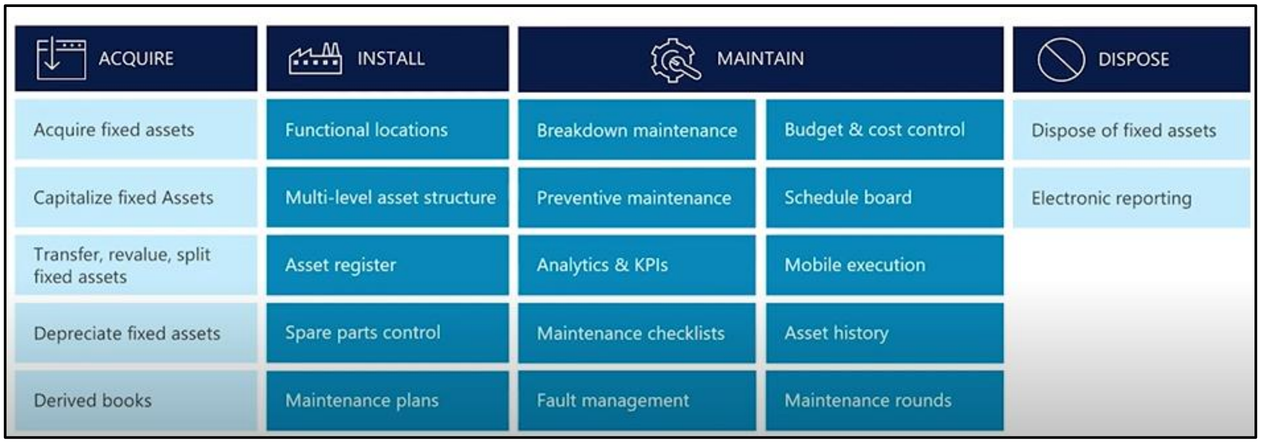


- **Work Description**



By Default, it will contain values stored on the Maintenance job type form. However, the same can be overwritten through this form and will flow to the Workorders.

All About Assets in D365(Fixed Assets+ Asset Management) is illustrated through the diagram below **(Source: Microsoft Learn)**



- **<u>Types of Maintenance</u>**

1. <u>Corrective / Reactive Maintenance</u>

Corrective Maintenance refers to unplanned maintenance activities triggered when an asset fails or shows signs of malfunction.

- **Maintenance Request**: A formal request initiated when an asset requires immediate attention or repairs due to breakdown or performance issues.

<u>In this document</u>, we will first configure the following **fundamental core components** for the **maintenance requests** discussed

a. Maintenance Request types

b. Lifecycle states

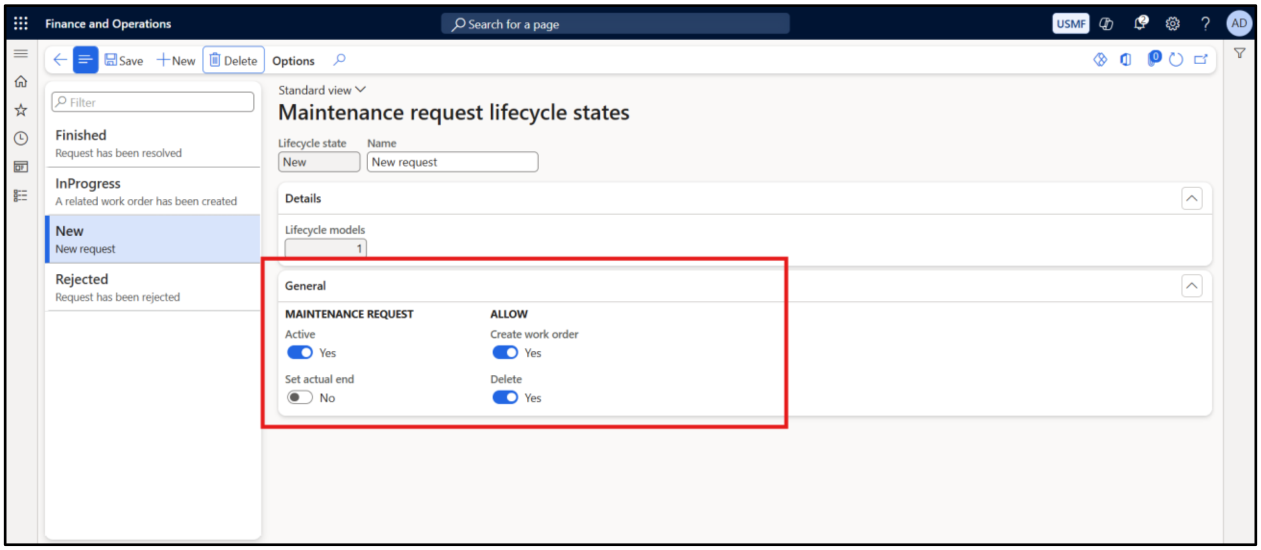
c. Lifecycle models

- **<u>Lifecycle states</u>**

As defined previously for location and assets, the lifecycle states definition remains the same for maintenance also. A **lifecycle state** represents a **stage in the life of maintenance request**, such as "New", "InProgress", "Finished", or "Rejected".

Each maintenance request moves through a **predefined sequence of lifecycle states**, and each state **governs what actions are allowed** (e.g., can it be edited, maintained, retired, etc.).

<u>Navigation:</u> Asset Management > Setup >Maintenance Request > Lifecycle states



<u>Details Tab</u>

The **Details** tab displays the number of **lifecycle models** that include this specific **lifecycle state**.

<u>General Settings</u>

Active: Indicates whether a maintenance request is considered active in this state.

Allow Create Work Order: Allows work orders to be created for maintenance requests that are in this state.

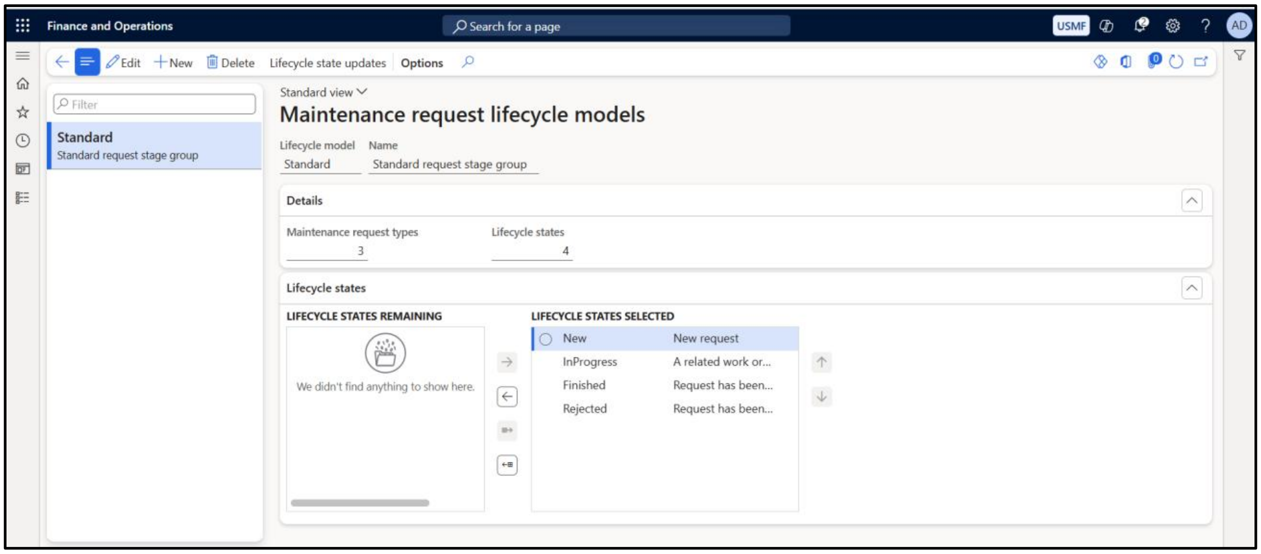
Delete: Permits deletion of maintenance requests when they are in this state.

- <u>**Lifecycle models**</u>

Represents a defined set of **lifecycle states** that a **maintenance request** can go through during its lifespan, such as “New,” “InProgress,” “Finished,” or “Rejected.”

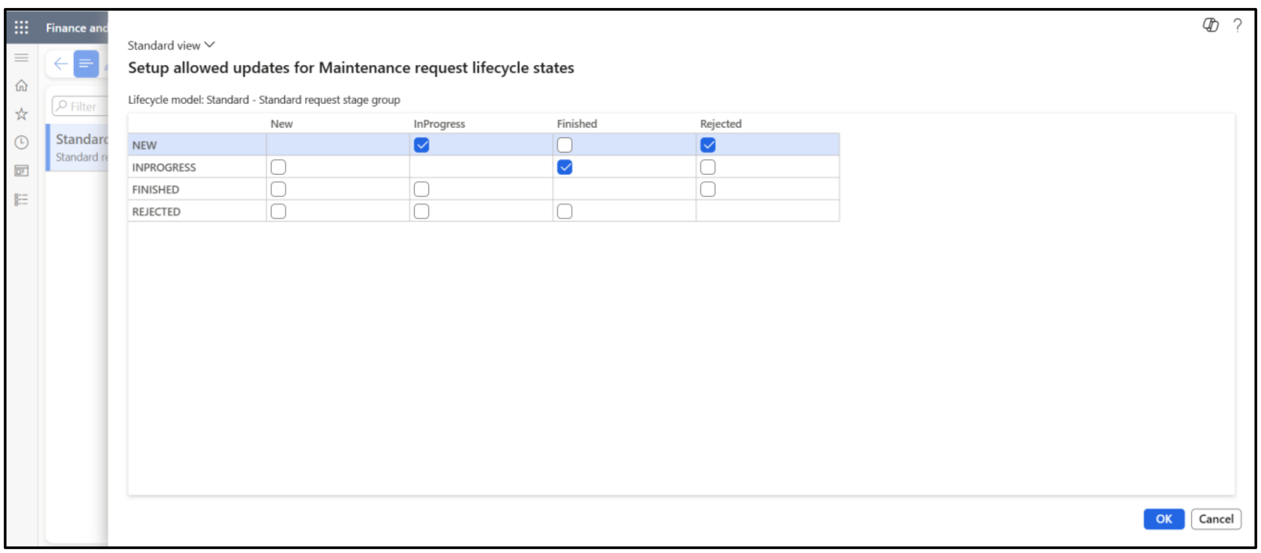
In simplified terms, it is a bundle of lifecycle states. All rules defined for each lifecycle states are applicable for the bundle.

<u>Navigation:</u> Asset Management > Setup > Maintenance Requests > Lifecycle models



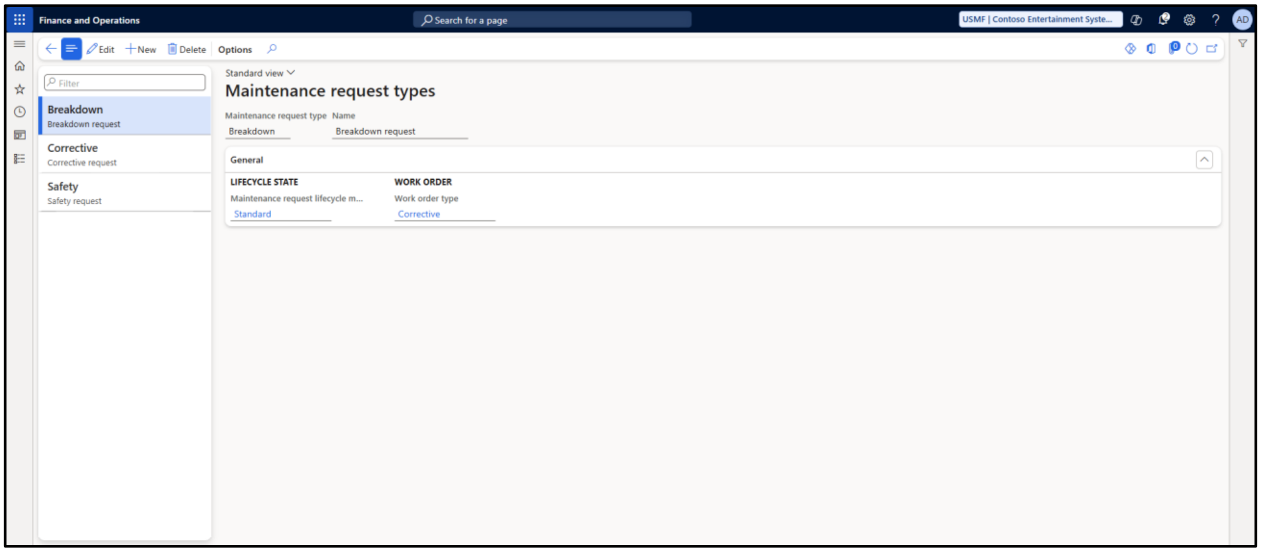
The Details tab represents the number of Maintenance Request types where this model is attached and the total number of lifecycle states which are a part of this model.

<u>Lifecycle status update</u>: This tab is used to define the movement from one lifecycle state to another.



- **<u>Maintenance Request types</u>**

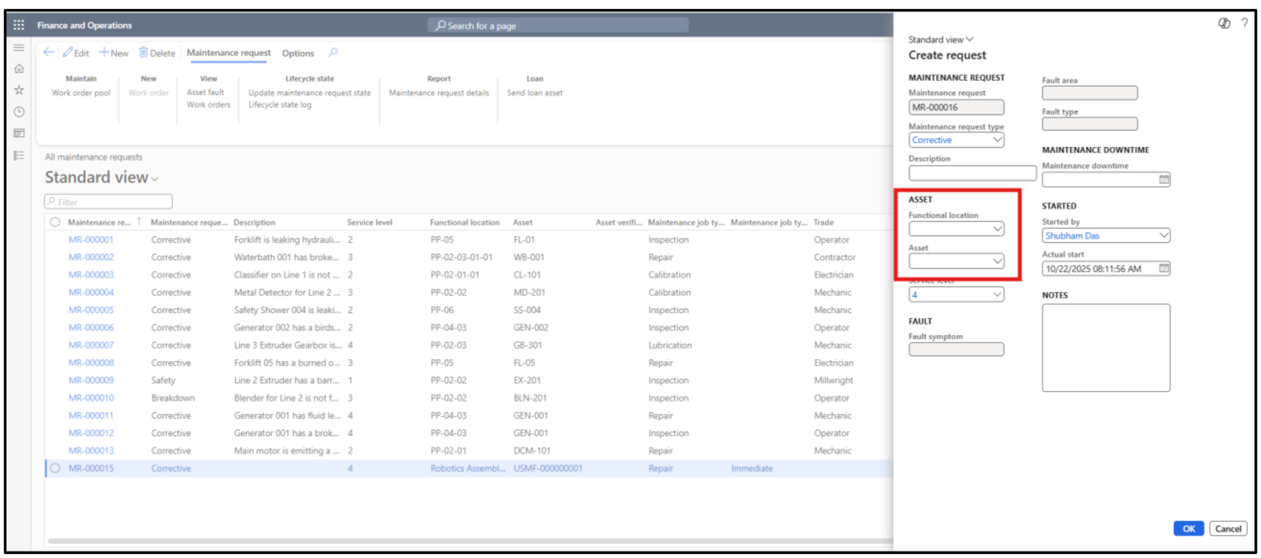
Maintenance Request Type is a category used to distinguish the intent of a maintenance request — for example, whether it is for a repair, inspection, replacement, or upgrade.**<u>Navigation:</u> Asset Management > Setup >Maintenance Request > Maintenance Request types**



- <u>**Creating a Maintenance Request**</u>

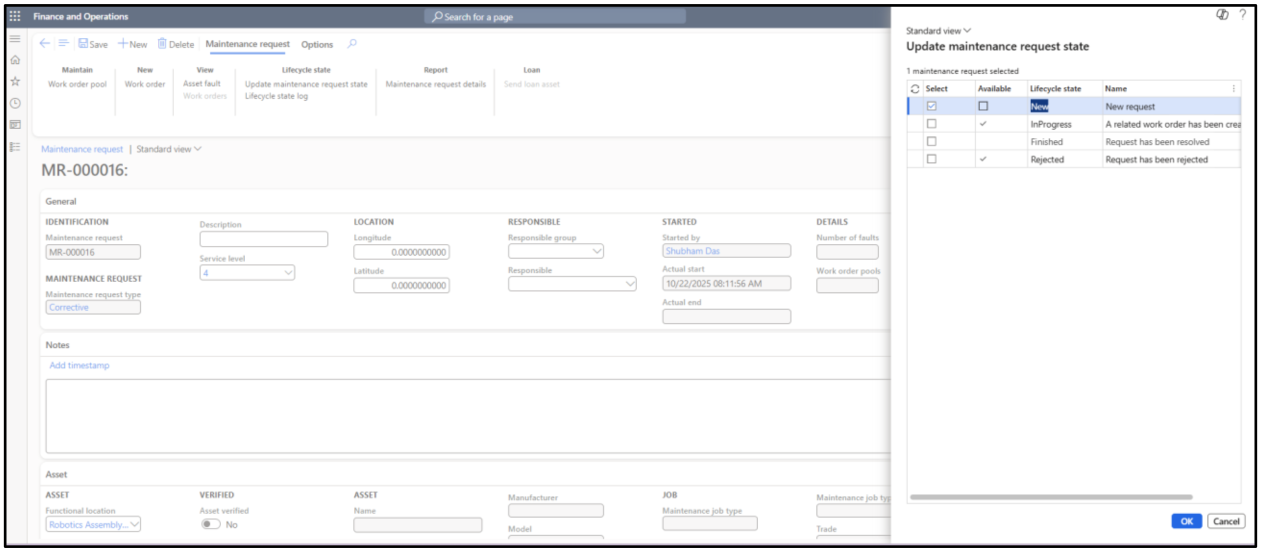
A maintenance request can be initiated either through the **Production Floor Execution** interface—where a worker submits a request in the event of a machine breakdown or repair need—or directly via the **Maintenance Request** form. In both cases, the request details are captured and reflected in the Maintenance Request form.

<u>Navigation:</u> Asset Management > Maintenance Request >All Maintenance Request



Enter the **Functional location** and/or **Asset**, then click **OK** to proceed. The maintenance request will progress through various lifecycle stages, eventually leading to the creation of related **work orders**. The monetary impact of these activities will be captured through **hour journals** and **item journals**, which are linked to a **project** for cost tracking and analysis.

These aspects will be explained in detail in the following section.

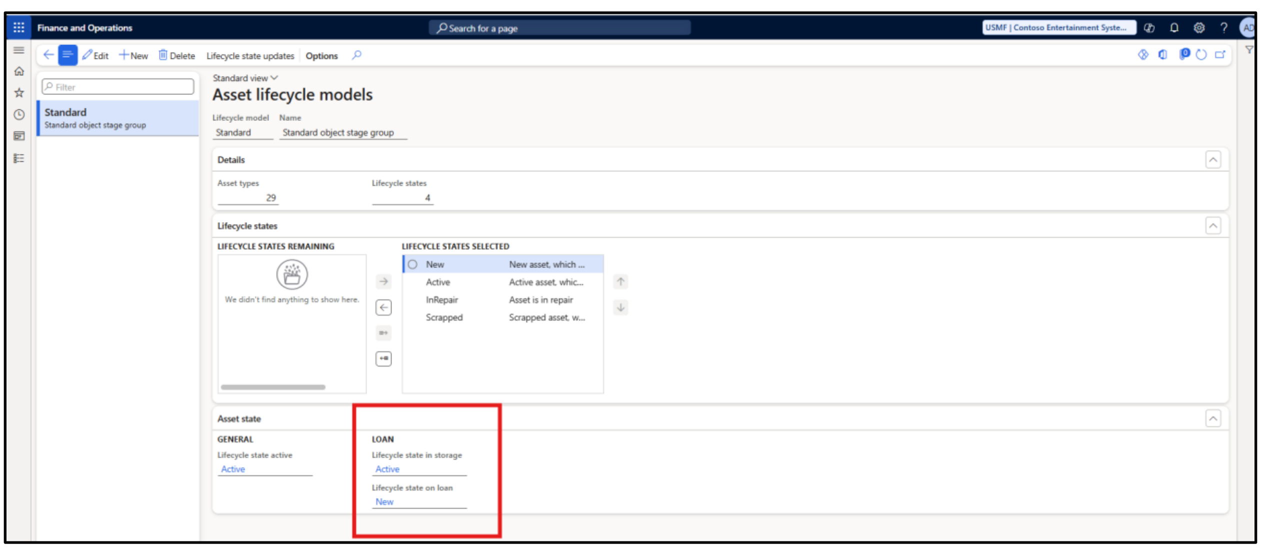


Concept of Asset Loan

If your company receives assets for repair or maintenance from internal departments or external customers — and if you also temporarily loan assets to those parties — the Asset Management module in Dynamics 365 can help you track these asset loans.

Assets can be loaned in connection with a Maintenance Request. To enable this functionality, the following condition must be met:

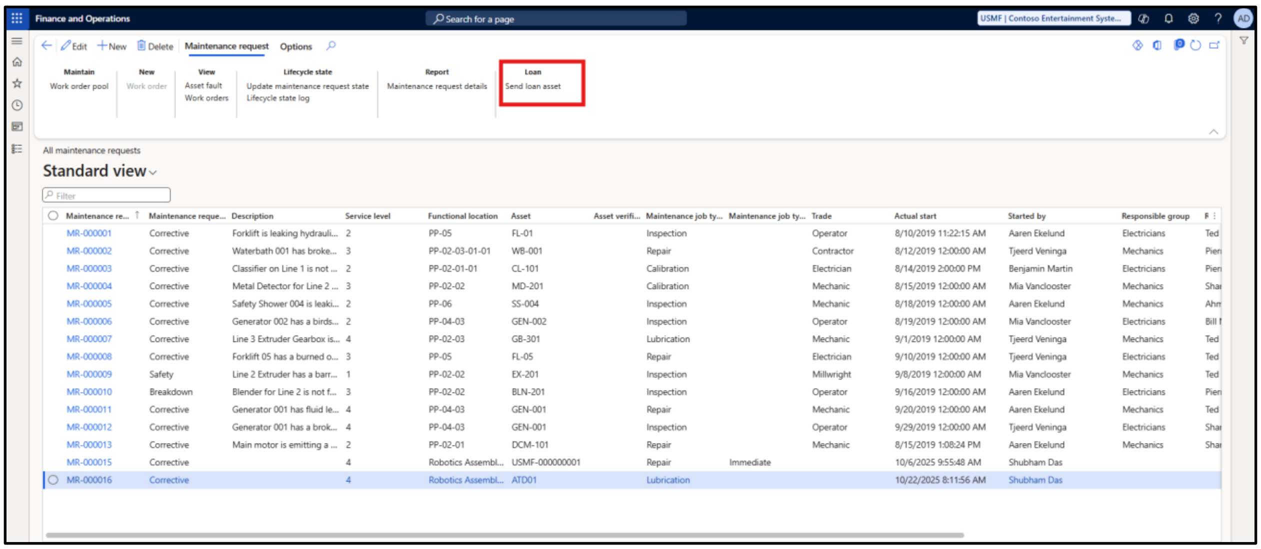
- The asset must be in a lifecycle state defined as **"In Storage"** within the Asset Lifecycle Model. Only assets in this state are eligible to be loaned



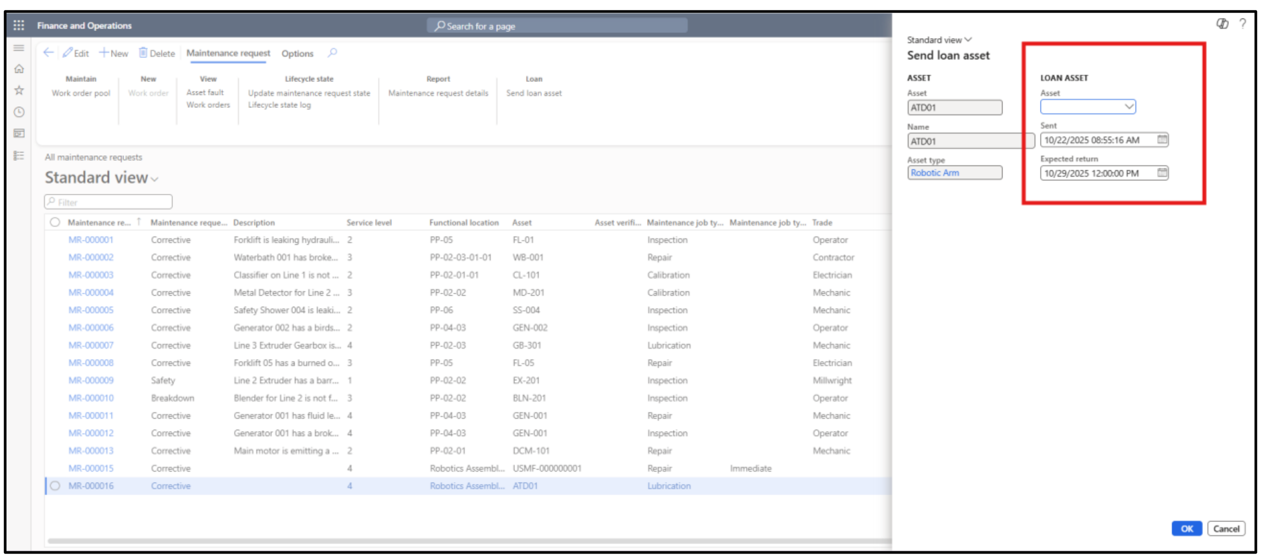
Only assets of the **same type** can be loaned against a maintenance request. The asset received for maintenance will have a specific asset type, and any asset provided as a loan must meet the following criteria:

- It must be in the **"In Storage"** lifecycle state
- It must be the **same** as the one being serviced

On the Maintenance Request form, select the Asset Loan as shown below:



The details of the asset to be loaned must be selected in relation to the asset received for maintenance



Once the asset is Loaned, it is no longer visible in the Active Asset list as the lifecycle stage of the asset would change in accordance with the setup explained above.

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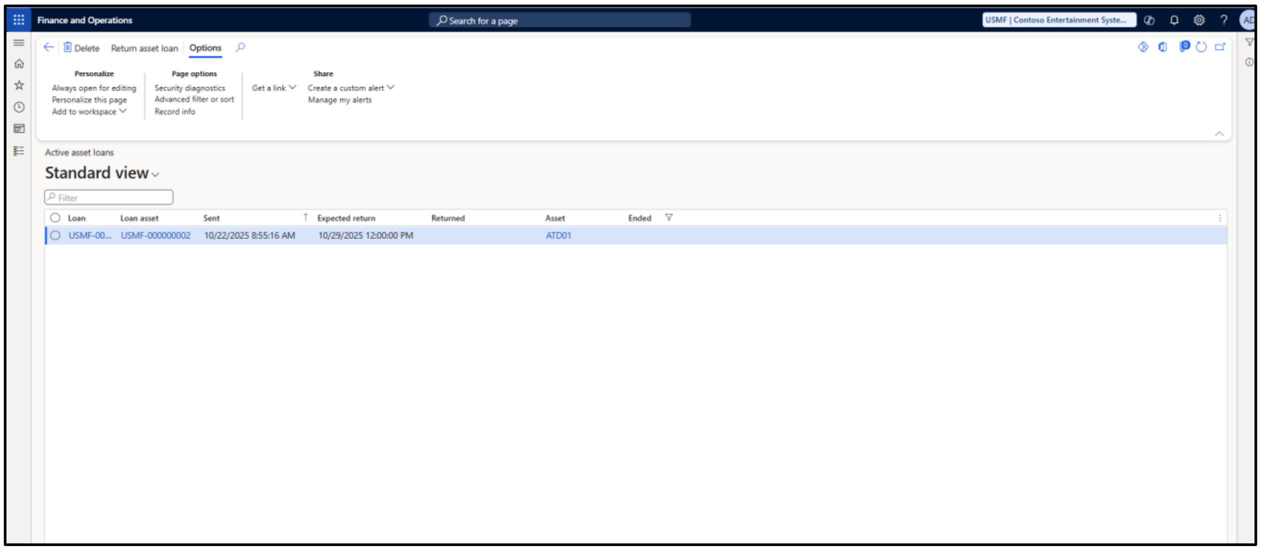
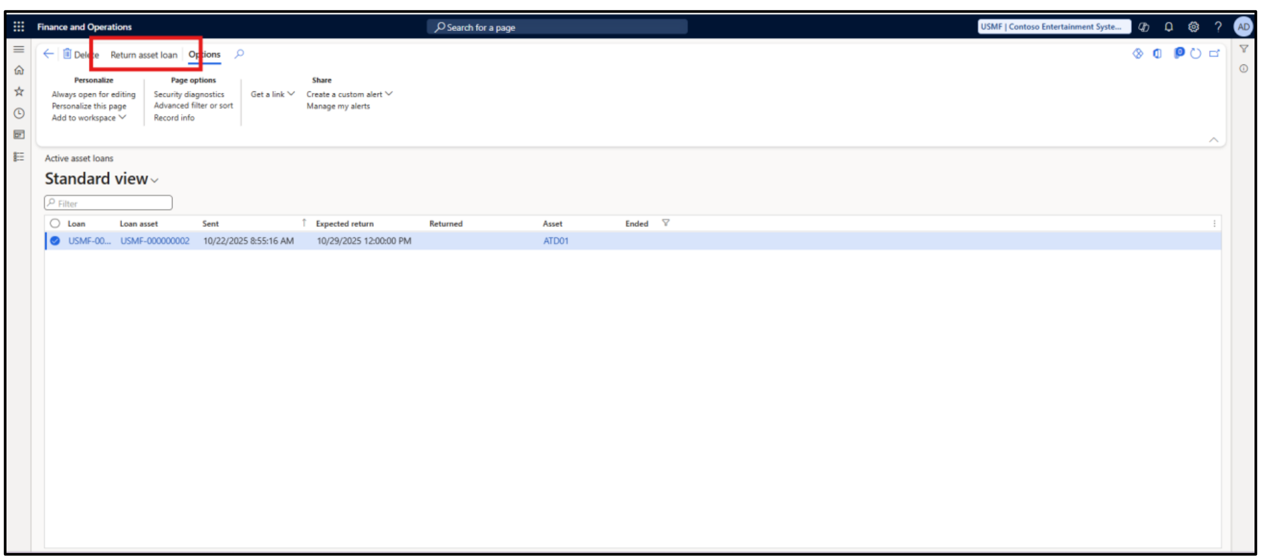
In our case we have set the same as “New” (this is a dummy data, and the recommendation would be to create a new state called “Loan”)

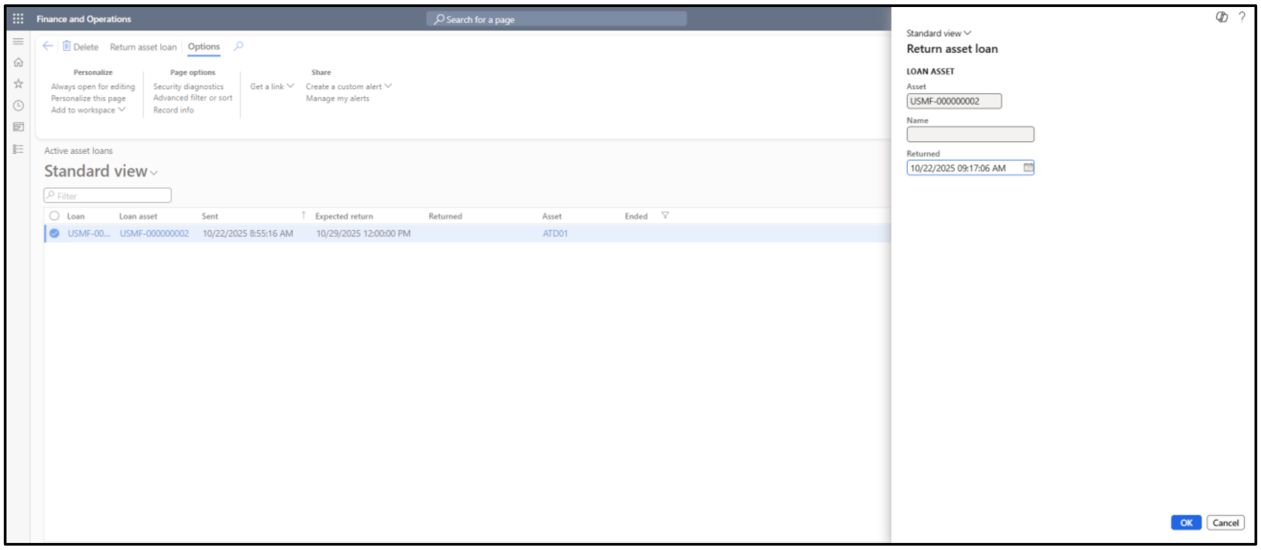
The details of the Loaned asset can be seen by Navigating to: **Asset Management > Asset Loan >Active Asset Loan**



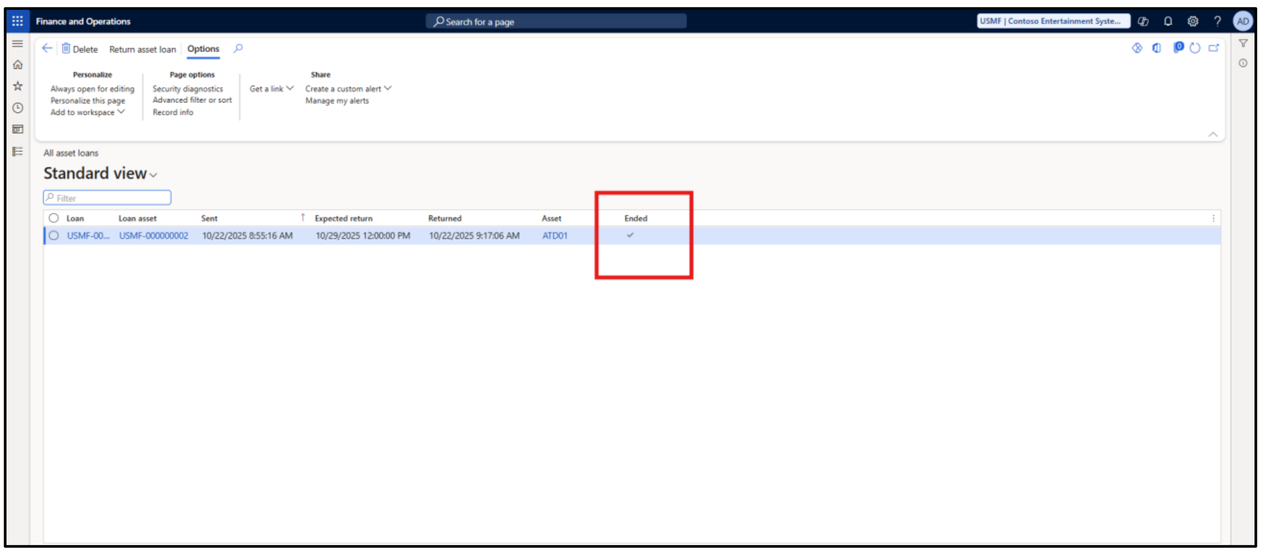
Once the Maintenance work is completed and the original asset of the customer is returned, the loaned asset can be brought back by selecting the asset on the above screen and selecting active asset loan.



The screen below will pop up



Once the asset is returned, the Ended checkbox will be enabled on the Asset loan and the same will again be visible on the Active Asset list.

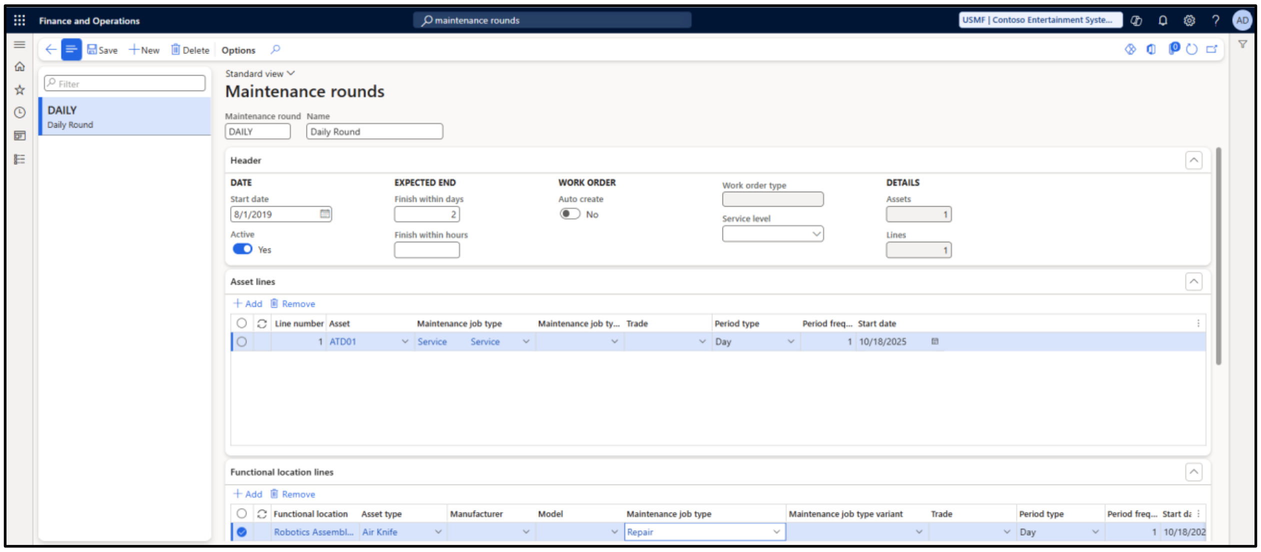


2. <u>Preventive / Predictive Maintenance</u>

Preventive Maintenance involves regularly scheduled tasks aimed at reducing the risk of failure and extending asset life.
Predictive Maintenance uses real-time data to anticipate failures before they occur.

- **Maintenance Rounds**: Routine maintenance activities conducted at scheduled intervals across multiple assets or asset types simultaneously.

<u>Navigation:</u> Asset Management > Setup > Preventive Maintenance > Maintenance Rounds



<u>Header</u>

This section contains the general set-up details for the maintenance round. It includes the start date**, the** expected duration **in days and hours, and specifies whether a** work order should be created automatically**. Additionally, the** service level **to be assigned to the generated work orders is defined here.**

Maintenance scheduling can be based on either functional location lines**,** asset lines**, or both. The related details are maintained within the same form.**

- **<u>Scheduling a Maintenance Round</u>**

Run a batch job/schedule the same by navigating to **Asset Management > Periodic > Preventive Maintenance > Maintenance Rounds**

Select the period for which Maintenance needs to be scheduled and execute the batch job.

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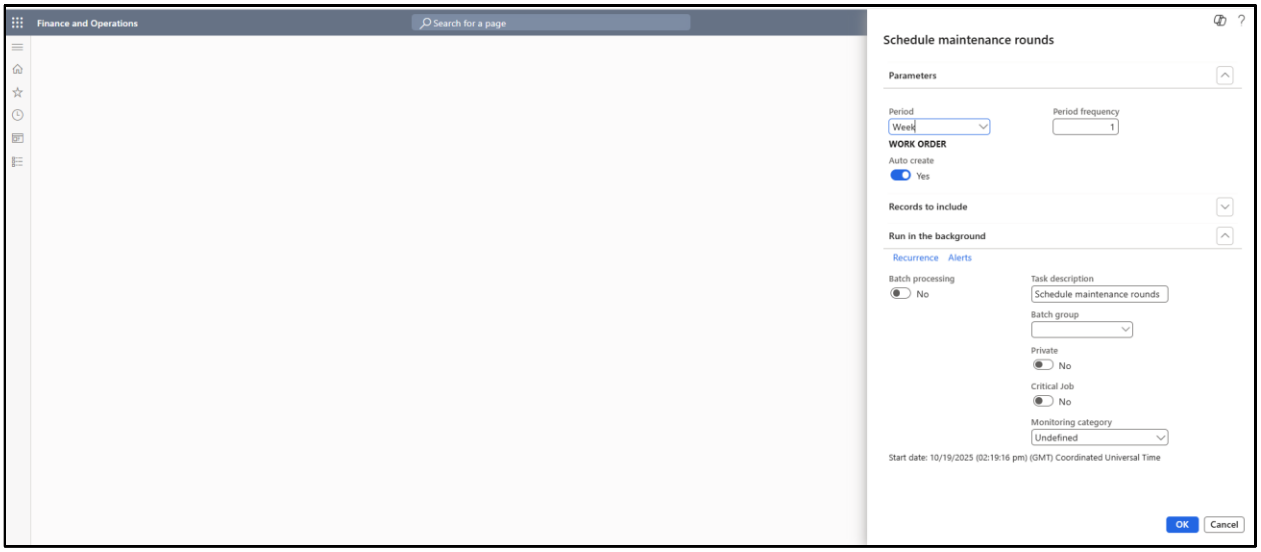
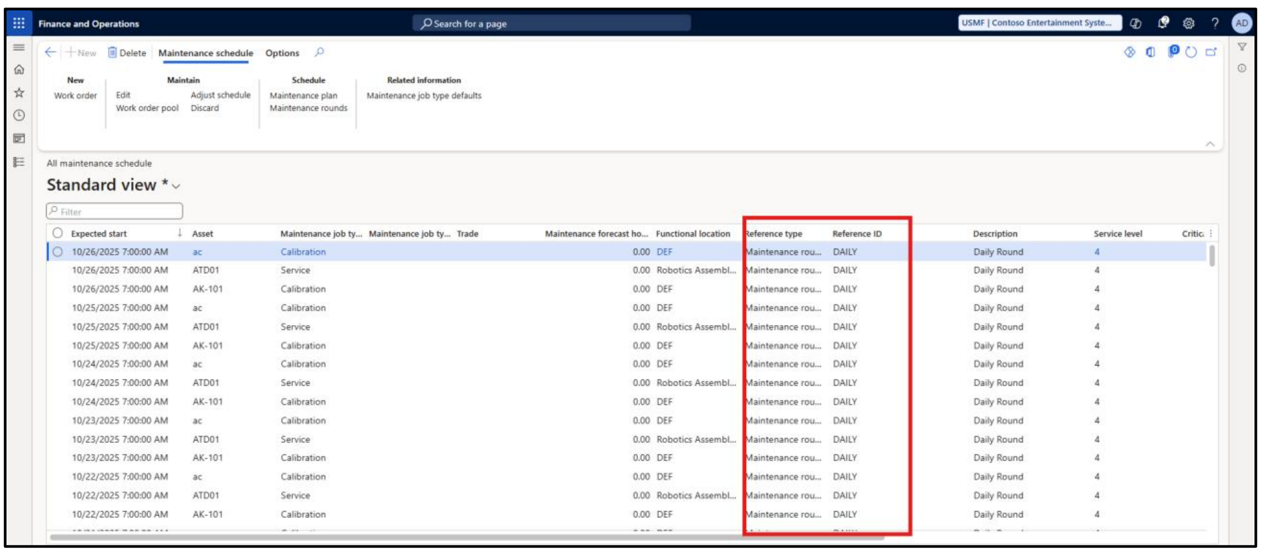


On the **Asset Management > Maintenance Schedule >All Maintenance Schedule,** the scheduled Maintenance activities related to maintenance round can be found.

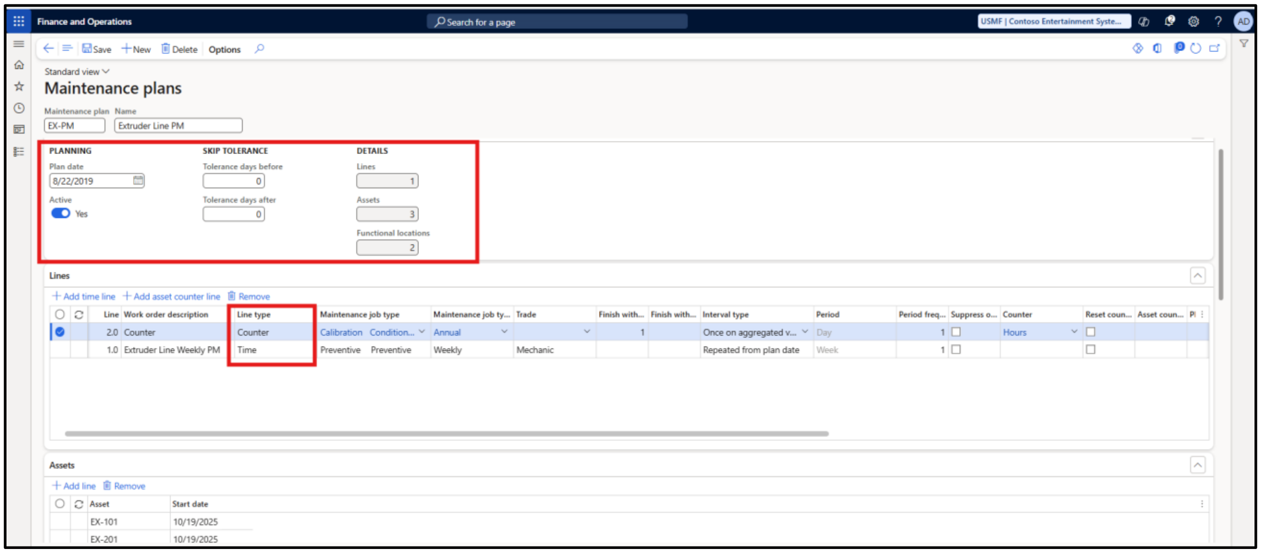


When to Use a Maintenance Round?

You have 10 machines across a production line. A technician does a **daily visual inspection** of all 10 machines in one go. You create a **maintenance round** that includes all 10 machines. Maintenance is **location- or route-based**, such as **daily inspections**.

- **Maintenance Plans**- Defined rules for scheduling preventive maintenance tasks, based on:
  - **Counter-based triggers** (e.g., runtime hours, cycles)
  - **Time-based intervals** (e.g., daily, monthly, annually)

<u>Navigation:</u> Asset Management > Setup > Preventive Maintenance > Maintenance Plans



The Planning section details the Planned start dates, any tolerances to be considered from the start date to schedule the maintenance and the details of the maintenance plan

<u>Lines</u>

Maintenance plan lines of type "Time" are used for recurring planned maintenance based on a fixed time interval. Maintenance plan lines of type "Counter" are used for planned maintenance or reactive maintenance based on asset counter registrations.

- <u>Some **important** field details</u>:

Work order description: The details here will flow to the workorders

Maintenance Job type, variants, and trade: Explained above

Finish within days and **finish within hours**: Expected end date in days and hours, respectively.

Interval type: Please refer to table mentioned at the end of document. This has been taken from https://learn.microsoft.com/en-us/dynamics365/supply-chain/asset-management/preventive-and-reactive-maintenance/maintenance-plans

Period: For Line type “Time” only. This refers to the periodic frequency of maintenance.

Period frequency: The number of times the line should be used for planning preventive maintenance jobs. Example: If you have created a line of type "Counter", and your counter is production quantity, and you insert the number "20000" in this field, new maintenance sequence lines are created during preventive maintenance scheduling every time you are expected to produce 20,000 more items

Suppress overlapping maintenance jobs check box relates to time-based as well as counter-based line types. Select the check box to delete maintenance schedule entries that are created on the same date.

The **Counter** field only relates to counter-based line types. Select the counter type to be used on the line. If a counter type is not active on a related asset, the maintenance plan line is omitted

Asset counter time fence in days-Applicable for counters. Consider counter readings from within the latest X days when calculating rate of change.

- **Maintenance Schedule**

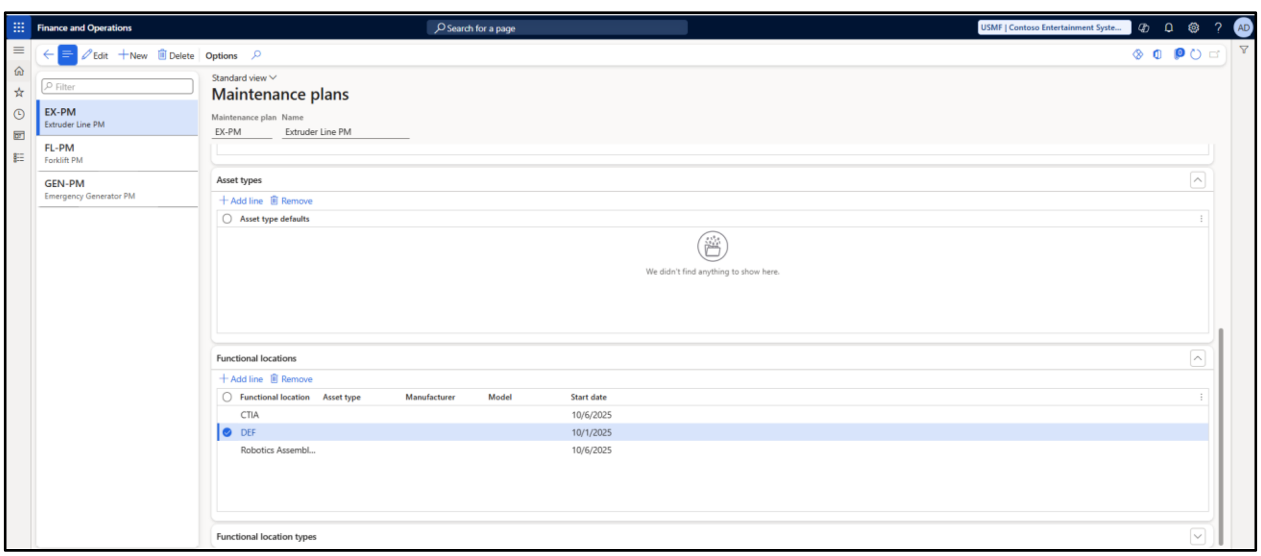
A **Maintenance Schedule** is a combination of both:

- **Preventive Maintenance** (Maintenance Rounds)
- **Preventive Maintenance** (Maintenance Plans)

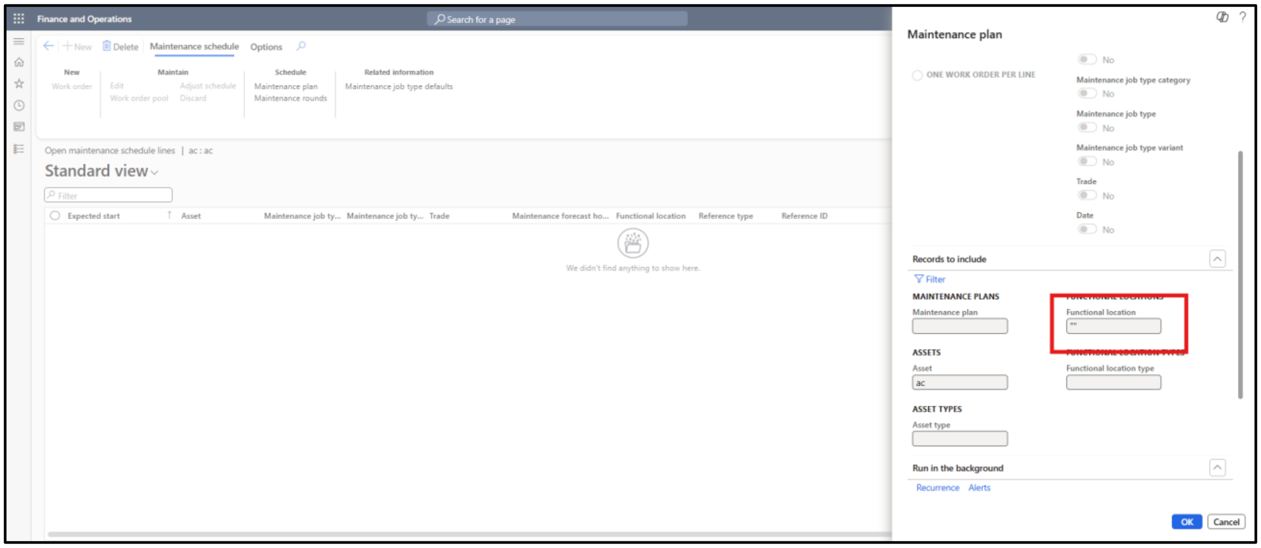
These schedules form the foundation for generating **Work Orders**, which are used to track:

- Labor time consumed
- Spare parts and materials used
  during the maintenance process.

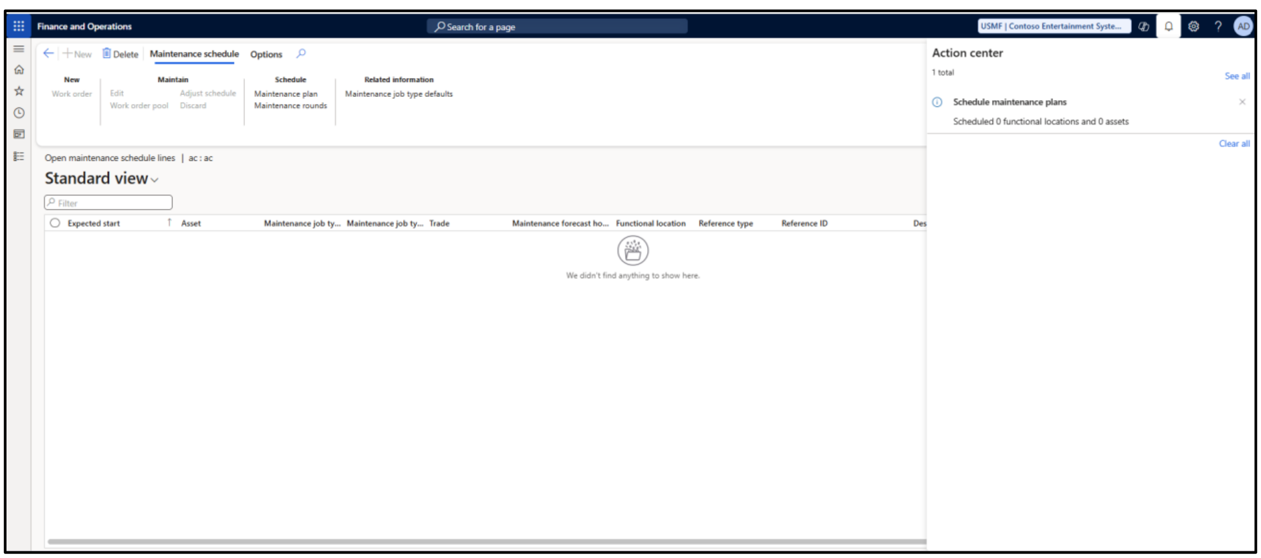
Case 1: Maintenance schedule for time based plan is tagged with only functional locations as shown below



If we try to execute the schedule by navigating to an asset which belongs to the functional location and then execute the schedule, the lines will not be created.

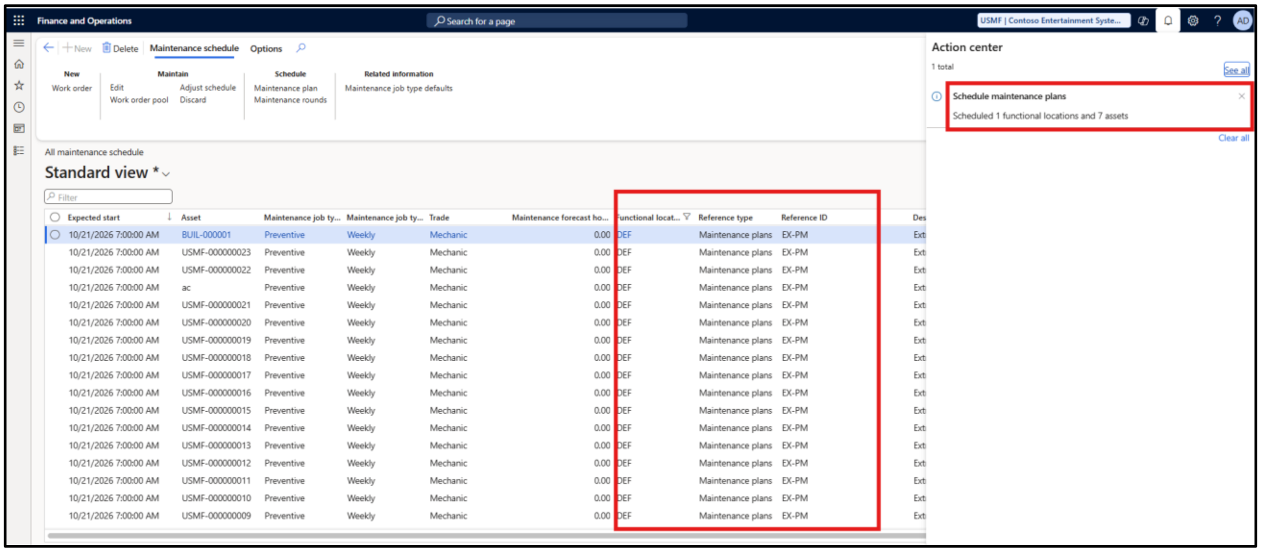


The reason for the same can be seen above. When the maintenance plan is executed through asset, the Functional location filter is blank by default. Hence, the system tries to see if the asset is a part of maintenance plan or not (through the asset tab). Since the asset is not found on the asset tab, hence no maintenance gets scheduled.



Conclusion: If plan has to run by asset, it is always recommended to maintain the asset on the asset tab in the maintenance plan.

Let us execute the same through functional location first and then later by clearing the value on the filter through the asset tab only and see the end result.



The maintenance schedule created using the counter based plan conditions also follows the same process.

In the next document we will deep dive into the Work orders and an end-to-end cycle for below process: -

1. Overview and setup of workorder and related setups
2. Maintenance request for work order execution
3. Maintenance schedule to Work order execution-Plan based
4. Maintenance schedule to Work Order execution- Counter based
5. Financial transactions related to Work Order- Excluding Project Setups

Interval Type (Maintenance Plans)

Interval type and description	**Line Type: Time**	**Line Type: Counter**
Interval type: Repeated from plan date The count starts from the plan date used. When you schedule maintenance plans, maintenance schedule lines are created when the interval is reached.	The **Plan date** on the maintenance plan line is used. If no plan date is selected on the line, the **Plan date** for the maintenance plan is used. Example: If the number "3" is inserted in the **Period frequency** field, and "Year" is selected in the **Period** field, a new maintenance schedule line will be created once every 3 years.	The **Plan date** for the maintenance plan is used. If the counter has been replaced, the latest replacement date is used as the plan date.
Interval type: Repeated from start date the count starts from the start date on the asset relation. The date is selected in the **All asset** detail's view > **Asset maintenance plans** Fast Tab > **Start date** field, or in the **All functional locations** detail's view > **Maintenance plans** Fast Tab > **Start date** field. When you schedule maintenance plans, a maintenance schedule line is created when the interval is reached.	The start date of the maintenance plan line on asset or functional location is used. If that field is blank, the **Plan date** for the maintenance plan is used.	The start date of the maintenance plan line on asset or functional location is used. If that field is blank, the **Plan date** for the maintenance plan is used.
Interval type: Repeated from last work order the count starts from the actual end date and time of the latest work order that was completed on the asset	The actual end date and time of the work order completed on the asset with that specific maintenance job	The actual end date and time of the work order completed on the asset *and* the maintenance job type / maintenance job type variant / trade

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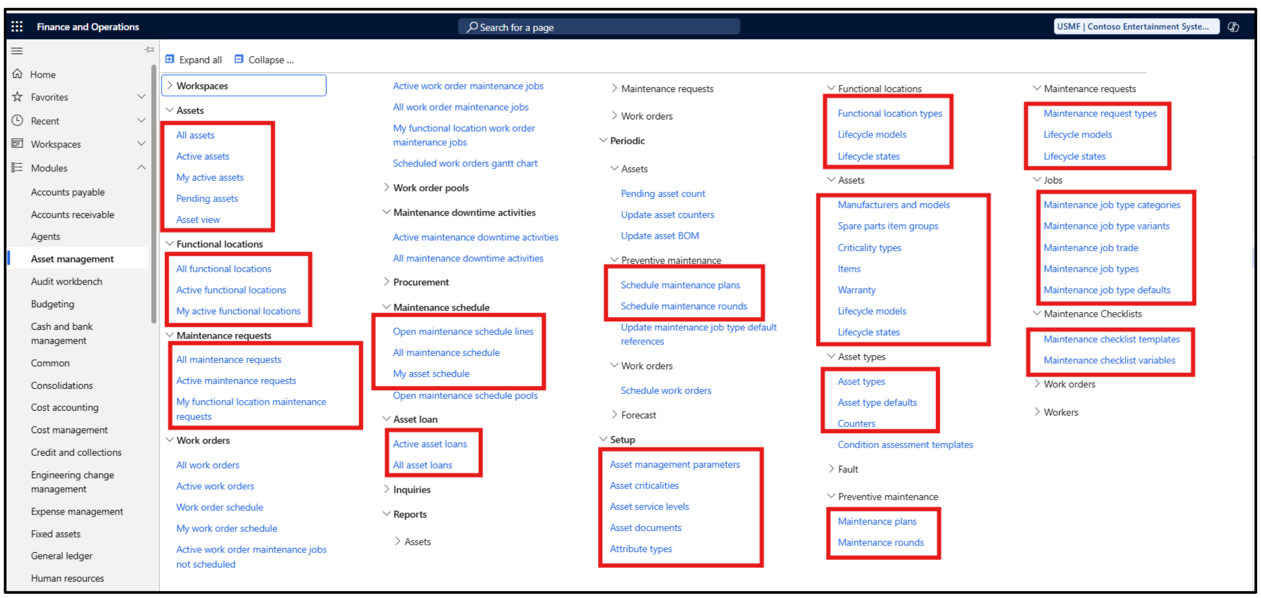
with that specific maintenance job type / maintenance job type variant / trade combination. That date and time are shown in the **Actual end** field in the **All work order** details view.	type / maintenance job type variant / trade combination. If no completed work order is found, one of the dates used in the "Repeated from start date" interval type is used instead.	combination. is used. If the end date and time was left blank on the work order, one of the dates used in the "Repeated from start date" interval type is used instead.
Interval type: Once from plan date See description for the "Repeated from plan date" interval type. Only difference is that this interval type is to be used only once.	See description for "Repeated from plan date" interval type. This interval is typically used for a one-time maintenance or service job.	See description for "Repeated from plan date" interval type above. This interval is typically used for a one-time maintenance or service job. **Note 1:** This interval type is only relevant if the counter is replaced every time, you carry out a maintenance or service job. If, for some reason, a counter has been replaced before the end of the planned interval, a new time is calculated for the job from the time of the counter replacement. **Note 2:** If the counter is replaced when completing the maintenance or service job, this interval type functions as the "Repeated from plan date" interval type.
Interval type: Once from start date See description for the "Repeated from start date" interval type. Only difference is that this interval type is to be used only once.	See description for "Repeated from start date" interval type. This interval is typically used for a one-time maintenance or service job.	See description for "Repeated from start date" interval type. This interval is typically used for a one-time maintenance or service job. **Note 1** above also applies to this interval type. **Note 3:** If the counter is replaced when completing the maintenance or service job, this interval type functions as the "Repeated from start date" interval type.
Interval type: Once reached above This interval type only relates to counters and is used for indicating an upper limit set up on the maintenance plan line. Maintenance schedule entries will have the expected start date and time of the counter registration, meaning these entries will be created with an expected start date equal to or earlier than the system date.	Not applicable	The counter interval indicates an upper limit. If that limit is exceeded when you create a counter registration, a maintenance schedule line is created when you schedule preventive maintenance.
Interval type: Once reached below This interval type only relates to counters and is used for indicating a lower limit set up on the maintenance plan line. Maintenance schedule entries will have the expected start date and time of the counter registration, meaning these entries will be created with an expected start date equal to or earlier than the system date.	Not applicable	The counter interval indicates a lower limit. If that limit is passed when you create a counter registration, a maintenance schedule line is created when you schedule preventive maintenance.
Interval type: Linked from start date This interval type only creates a maintenance schedule line once. A maintenance plan can contain more maintenance plan lines using this interval type, and those lines are linked. Typically, you will create a maintenance plan that contains lines of only this interval type. Maintenance schedule lines are created by identifying the maintenance plan line that has the first expected start date and time.	See description for "Once from start date" above. Example: You create two lines in a maintenance plan for a service job on a car: one time-based line with a 1-year period, and one counter-based line with a 25,000 km limit. A maintenance schedule line is created for the limit that is reached first. For this line type you create the line with the 1-year period.	See description for "Once from start date" above. Example: You create two lines in a maintenance plan for a service job on a car: one time-based line with a 1-year period, and one counter-based line with a 25,000 km limit. A maintenance schedule line is created for the limit that is reached first. For this line type, you create the line with the 25,000 km limit. Example creating two counter lines: You can also set up a maintenance plan with two linked, counter-based lines in which the first line has a limit of 10,000 items quantity produced, and the second line relates to the machine or work center requiring service after running 3,000 hours.
Interval type: Linked from last work order This interval type only creates a maintenance schedule line once, based on the latest work order completed on the asset that matches the maintenance job type, maintenance job type variant, or trade combination. A maintenance plan can contain more lines using this interval type, and those lines are linked. Typically, you will create a maintenance plan that contains maintenance plan lines of only this interval type. Maintenance schedule lines are created by identifying the maintenance plan line that has the first expected start date and time.	This interval type works as "Linked from start date" described above. Only difference is the date on which the interval type is based. The date used is the actual date and time on the latest work order completed on the asset *and* the maintenance job type / maintenance job type variant / trade combination.	This interval type works as "Linked from start date" described above. Only difference is the date on which the interval type is based. The date used is the actual date and time on the latest work order completed on the asset *and* the maintenance job type / maintenance job type variant / trade combination.

Interval type: Repeated on aggregated value (Counter only) When the maintenance plan is run, a scheduled maintenance line will be created each time that the accumulated value for an asset counter reaches the period frequency or an even multiple of the period frequency. (The period frequency is defined on the maintenance plan line.) For more information about how to enable and use this functionality, see the [Counter-based maintenance enhancements]() section later in this article.	Not applicable	**Example:** An hour counter is set up for asset AK-101. An asset plan line is also set up for the asset. The interval type of this line is *Repeated on aggregated value (Counter only)*, and the period frequency is *1000*. When the maintenance plan is run, a scheduled maintenance line will be generated when the aggregated value for the counter exceeds 1,000 hours. Then, when the aggregated value for the counter exceeds 2,000 hours, another scheduled maintenance line will be generated, and so on for every additional 1,000 hours.
Interval type: Once on aggregated value (Counter only) When the maintenance plan is run, a scheduled maintenance line will be created when the accumulated value for an asset counter reaches the period frequency that is defined on the maintenance plan line. For more information about how to enable and use this functionality, see the [Counter-based maintenance enhancements]() section.	Not applicable	**Example:** An hour counter is set up for asset AK-101. An asset plan line is also set up for the asset. The interval type of this line is *Once on aggregated value (Counter only)*, and the period frequency is *1000*. When the maintenance plan is run, a scheduled maintenance line will be generated when the aggregated value for the counter exceeds 1,000 hours.

In the three parts we have covered the aspects marked in Red in this document.



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